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ACTIVITY 4 — PROCESS MANAGEMENT

Application Activities Based on Module 4

Activity Overview

This activity applies the concepts discussed in Module 4: Process Management. Complete the tasks using the module as your primary reference. Answers should use correct process-management terminology and should show the reasoning behind each conclusion.

Learning Targets

1. Identify process states and explain process-state transitions.
2. Interpret the contents and role of the Process Control Block (PCB).
3. Classify processes according to scheduling queues and scheduler types.
4. Explain context switching, process creation, and termination using realistic scenarios.
5. Analyze process isolation and interprocess communication in a multiprocess system.

General Instructions

6. Answer all parts of the activity.
7. Use complete sentences for explanation questions unless a table or diagram is specifically requested.
8. Base answers on concepts from Module 4. You may observe your own computer only when a task explicitly asks you to do so.
9. Do not simply copy definitions. Apply each concept to the given scenario.
10. Submit the completed worksheet/document using the file naming format required by your instructor.

Filename:Activity4_Lastname_Firstname.docx

Part A — Process State Analysis

For each scenario, identify the most appropriate process state: New, Ready, Running, Waiting, or Terminated. Then explain why the process is in that state.

No.	Scenario	Process State	Explanation
1	The OS has just created a process and is preparing its management information.	New	The process is in the New state because the Operating System has just created a process and is setting up its



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			management information.
2	A process is in main memory and is waiting for the CPU scheduler to select it.	Ready	The process is in the Ready state because it is already in main memory and can run, but another process currently has the CPU.
3	The CPU is currently executing the process instructions.	Running	The process is in the Running state because the CPU is currently executing its instructions.
4	A process requested a disk read and cannot continue until the data arrives.	Waiting	The process is in the Waiting state because it cannot continue until the requested I/O operation is completed and the data arrives.
5	The process has completed its last instruction and its resources are being released.	Terminated	The process is in the Terminated state because the process has completed its last instruction and its resources are already being released.
6	A running process loses the CPU because its time slice has expired. What state will it normally enter next?	Ready	The state of the process that it will normally enter next is the Ready state, because it has not finished executing yet, the running process just loses the CPU, and it will return to the ready queue to wait for another CPU turn.
7	An I/O request completes for a waiting process. What state will it normally enter next?	Ready	The state of the process that it will normally enter next is from the Waiting state to the Ready state, because the required I/O event has already been completed and it will just wait for the CPU

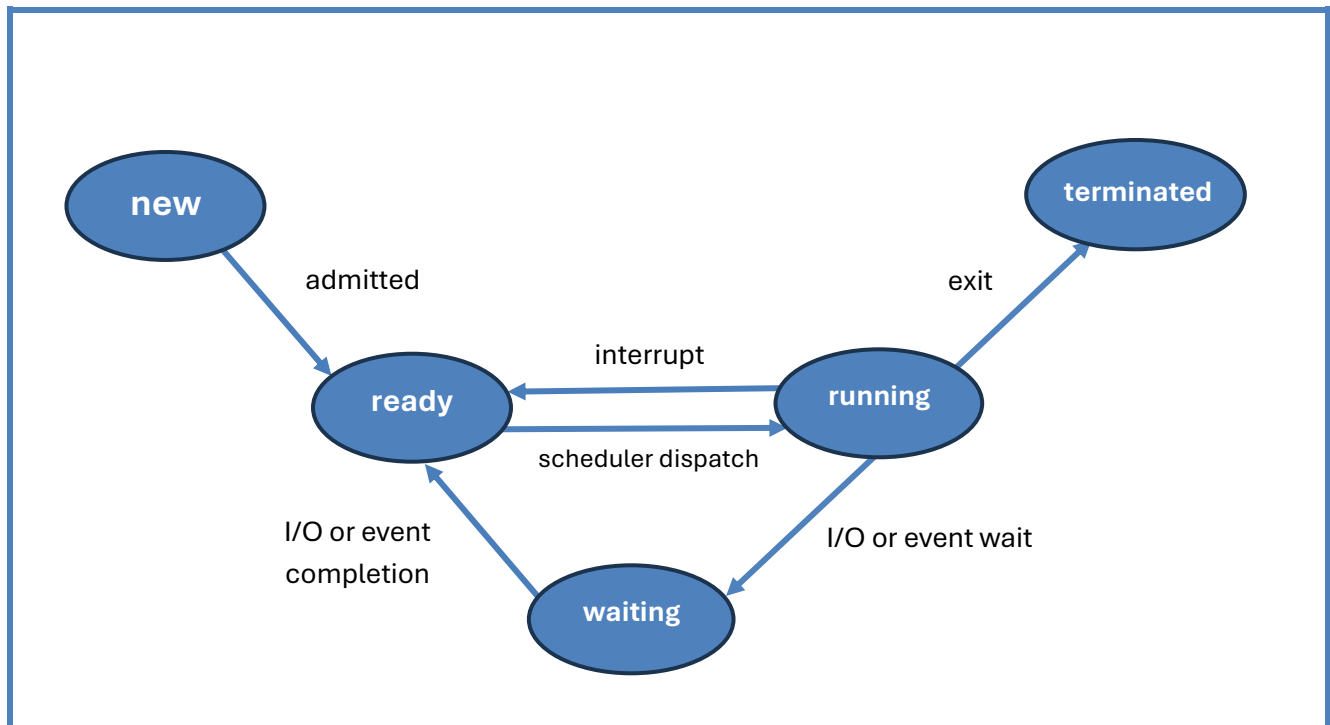


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			scheduler to give it CPU time.
8	The scheduler chooses a process from the ready queue. What state will it enter?	Running	The state of the process that it will normally enter next is the Running state, because the CPU scheduler has already selected it and allocated CPU time to it.

Draw the five-state process diagram from memory and label these transitions: admitted, scheduler dispatch, interrupt, I/O or event wait, I/O or event completion, and exit.

Five-State Process Diagram





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Part B — PCB Investigation

Match each process-management need with the PCB information that would be most useful. Use the following choices: Process State, Program Counter, CPU Registers, CPU Scheduling Information, Memory-Management Information, Accounting Information, and I/O Status Information.

No.	Situation	PCB Information
1	The OS must know the next instruction to execute when a process resumes.	Program counter
2	The OS must restore values that were previously in the processor registers.	CPU register
3	The OS needs to know the process priority and queue-related data.	CPU scheduling information
4	The OS must determine what memory is allocated to the process.	Memory-management information
5	The OS needs to know which files are open and which I/O devices are assigned.	I/O status information
6	The OS wants to record how much CPU time the process has already used.	Accounting information
7	The OS needs to know whether the process is ready, running, or waiting.	Process state

Short Analysis: Why is the PCB essential during a context switch? Explain in 4–6 sentences.

The PCB is essential during a context switch because it stores important information about a process. When the operating system interrupts a process, it saves the process's current program counter and CPU register values in its PCB. The operating system can then load the saved information of another process so that it can continue its execution. When the first process runs again, its saved information lets it continue from where it left off. Therefore, the PCB helps the operating system switch between processes without losing their execution information.



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Part C — Scheduling Queues and Schedulers

Complete the following classification tasks.

Item	Answer
1. Queue that contains all processes in the system	Job Queue
2. Queue that contains processes in main memory waiting for CPU execution	Ready Queue
3. Queue that contains processes waiting for an I/O device	Device Queue
4. Scheduler that selects the next ready process and allocates the CPU	Short-term scheduler (CPU scheduler)
5. Scheduler that selects which processes are brought into the ready queue	Long-term scheduler (job scheduler)
6. Scheduler that may remove a process from memory and later restore it	Medium-term scheduler
7. Term for temporarily moving a process from memory to disk	Swapping
8. Process type with many short CPU bursts and more time spent doing I/O	I/O-bound process
9. Process type with longer computation periods and fewer long CPU bursts	CPU-bound process

Scenario Analysis: A computer is running a video encoder, a file downloader, a word processor, and a backup utility. Identify one task that is likely CPU-bound and one that is likely I/O-bound. Justify each choice in 2–3 sentences.

CPU-bound: Video encoding is typically a CPU-bound process due to the high level of computational intensity it requires. As a result, the encoder often experiences extended CPU bursts while performing complex calculations. This heavy computational demand can lead to increased processing times during the encoding phase.

I/O-bound: The file-downloader is likely I/O-bound because it spends significant time waiting for network or storage operations. This means it performs more I/O activity compared with computation.



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Part D — Context Switch and Process Operations

Read the scenario and answer the questions.

Scenario — Process A is running. Its time slice expires, so the OS switches to Process B. Later, Process B requests an I/O operation. Meanwhile, Process A is selected again. Process A then creates a child process to perform another task.

11. What information about Process A must be saved before Process B can run?

Answer: The information that must be saved is **Process A's current process state, program counter, and CPU register values** before Process B can run.

12. Which operating-system data structure stores the saved process context?

Answer: The operating system data structure that stores the saved process context is the **Process Control Block (PCB)**.

13. Why is context-switch time considered overhead?

Answer: Context-switch time is considered overhead because the CPU is performing process-management work, such as saving and loading process information, instead of performing useful application work.

14. When Process B requests I/O and cannot continue, what process state does it enter?

Answer: Process B enters the **Waiting state** because it must wait for the requested I/O operation to complete.

15. What relationship exists between Process A and the newly created process?

Answer: The newly created process is the **child process**, while Process A is the **parent process**.

16. What UNIX system call creates a new process?

Answer: `fork()`

17. What UNIX system call can replace a process memory space with a new program?

Answer: `exec()`

18. What system call is used when a process finishes normally in UNIX-like systems?

Answer: `exit()`

19. What call allows a parent to receive termination status from a child?

Answer: `wait()`



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Explain the full sequence of the context switch from Process A to Process B in 5–7 sentences.

When Process A is running and its time slice expires, the operating system interrupts Process A. The OS saves Process A's current state, including its program counter and CPU register values, in Process A's PCB. The scheduler then selects Process B from the ready queue. The OS loads Process B's previously saved process information from its PCB. Process B can then continue execution using its saved state. The CPU is now executing Process B, while Process A remains ready to receive another CPU turn later. The time spent saving and loading these process states is considered context-switch overhead.

Part E — Multiprocess Browser and IPC Case Study

A browser has one main browser process and several renderer processes. One renderer becomes unresponsive because a web page executes faulty code. At the same time, another renderer must send data to a separate helper process.

20. Why can a multiprocess browser remain more stable than a single-process browser when one page fails?

Answer: A multi-process browser can remain more stable because different web pages can run in separate renderer processes. If one renderer has a problem, the failure is less likely to crash the entire browser because the processes are separated.

21. What is the role of the browser process according to Module 4?

Answer: The browser process manages the user interface, disk I/O, and network I/O.

22. What is the role of a renderer process?

Answer: A renderer process renders web pages and handles content such as HTML and JavaScript. Separate renderer processes can also help isolate websites.

23. How does sandboxing help reduce the effect of security exploits?

Answer: Sandboxing restricts a renderer's direct access to resources such as the disk and network. This limits what a compromised renderer can access and can reduce the impact of some security exploits.

24. Why do cooperating processes need IPC?

Answer: Cooperating processes need IPC so they can exchange information and coordinate their work.

25. Name and briefly explain the two IPC models in Module 4.

Answer:

- **Shared Memory** - Processes communicate through a common memory region that both processes can access.



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- **Message Passing** – Processes communicate by sending and receiving messages through operating-system-supported communication mechanisms.

26. Give one example in which shared memory would be useful.

Answer: Shared memory would be useful when two cooperating applications exchange data through a shared buffer.

27. Give one example in which message passing would be useful.

Answer: Message passing would be useful when one process sends a request to another process and the second process sends a response back without directly sharing an address space.

Part F — Real-System Observation

Use Windows Task Manager, Linux System Monitor, or an equivalent process-monitoring tool. Do not end system processes. Observe only.

28. Open at least three user applications.
29. Open the operating system process-monitoring tool.
30. Record at least five visible processes, including at least one background/system process if identifiable.
31. Record the visible CPU and memory activity for each process.
32. Choose one process and explain whether its observed behavior appears more CPU-oriented, I/O-oriented, or cannot be determined from the available information.
33. Write a 150–200 word reflection explaining how process management allows several applications to appear to run at the same time.

Process / Application	User or Background/System	CPU Activity	Memory Use	Observation
Microsoft Edge	User	0.2%	440.3 MB	The CPU usage changes quickly, memory usage also increases and decreases. While using Efficiency mode, I observed that Edge's memory usage sometimes became higher than Spotify's, but the memory usage later changed and



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				Spotify became higher.
Microsoft Word	User	0.4%	73.7 MB	Its memory usage decreased significantly compared with my earlier observation. Unlike some other processes, its memory usage does not change as quickly and sometimes decreases.
Spotify Widget	User	0%	258.6 MB	There was a time when Spotify's memory usage was higher than Edge's, but later Edge used more memory than Spotify.
Steam Client WebHelper (7)	Background process	0.4%	54.3 MB	It does not use much CPU compared with the other observed processes, and its memory usage does not increase much.
WebView2 Manager (9)	Background process	0.4%	174.6 MB	Its memory usage increases and decreases noticeably. I observed that it changes while the process is running in Efficiency mode.



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Figure 1. Application Process and Background Processes Observation

Processes						
Name	Status	21% CPU	97% Memory	10% Disk	0% Network	
Apps (4)						
> Task Manager		4.4%	68.9 MB	0 MB/s	0 Mbps	
> Spotify Widget (12)		0%	258.6 MB	0.2 MB/s	0 Mbps	
> Microsoft Word		0.4%	73.7 MB	0 MB/s	0 Mbps	
> Microsoft Edge (13)		0.2%	440.3 MB	0.1 MB/s	0 Mbps	
> WidgetsPlatformRuntime		0%	0.8 MB	0 MB/s	0 Mbps	
> WebView2 Manager (9)		0.4%	174.6 MB	0 MB/s	0 Mbps	
> WebView2 Manager (6)		0.2%	12.9 MB	0.1 MB/s	0 Mbps	
Superhuman Web UI		0%	4.6 MB	0 MB/s	0 Mbps	
> Steam Client WebHelper (7)		0.4%	54.3 MB	0.1 MB/s	0 Mbps	

32. Process Activity Analysis (CPU-bound or I/O-bound)

I chose **Microsoft Edge**. Its CPU usage changed quickly, increasing from around 0.2% to about 6.0% and then decreasing and increasing again. However, this observation alone is not enough to determine whether Edge is CPU-bound or I/O-bound because the available CPU and memory information does not show how much time the process spends performing computation or waiting for I/O operations. Therefore, its behavior cannot be determined from the available information.

33. Reflection (150-200 words)

Observing the process in Windows Task Manager helped me understand how process management works in an actual computer. I noticed that the CPU and memory usage of applications such as Microsoft Edge, Microsoft Word, and Spotify changed over time. Some processes changed quickly, while others remained more stable. I also observed background processes such as Steam Client WebHelper and WebView2 Manager, which were running even though I was not directly using them. This showed me that many processes can be active at the same time and that they do not always use the same amount of CPU and memory. I also



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learned that CPU-bound or I/O-bound. Process management allows the operating system to organize these processes and share CPU time among them. Through process states, queues, scheduling, and context switching, the operating system allows different applications to make progress without requiring the user to manually manage each process. This activity helped me understand how process management works in real computer use rather than only in theory.

Required Output

- Completed Parts A–F.
- One hand-drawn or digitally created five-state process diagram.
- Completed process-observation table.
- 150–200 word reflection for Part F.

Scoring Guide

Criterion	Description	Points
Process State Analysis	Correct identification and explanation of states and transitions.	20
PCB and Context Switch	Accurate use of PCB fields and context-switch concepts.	20
Scheduling Concepts	Correct classification of queues, schedulers, CPU-bound and I/O-bound processes.	20
Process Operations and IPC	Correct application of creation, termination, multiprocess, and IPC concepts.	20
System Observation and Reflection	Complete observation, relevant evidence, and clear reflection.	15
Organization and Presentation	Complete, readable, and properly organized submission.	5
TOTAL		100